

Measuring the Effect of Iran War Risk on Global Financial Markets

FRE-GY 7871 A · NLP and the Investment Process · Fall 2026

0. The problem and the method

War risk is unobservable: it is easy to say *when* war news broke, much harder to say by how much it moved the odds of war. Rigobon and Sack (2003) identify the effect from a shift in *variance* rather than the level of the factor — name a set of days *H* on which war-related news was unusually intense, assume every other driver of asset prices was no more volatile than usual on those days, and the shift in the covariance matrix of returns between *H* and a comparison set *L* is attributable to the war factor alone, without ever measuring the factor's level. Full derivation and the estimator implementation are in `src/heteroskedasticity.py`.

Every trading day carries a binary flag, `war_news_flag` — **1 if it is a high-news day, 0 otherwise** — built from an NLP measure of GDELT news coverage (scripts 03/04). This flag is what feeds the heteroskedasticity estimator; the estimator does not need to know the *direction* of the news, only that variance was elevated on those days. `data/interim/event_days.csv` carries this flag for all 173 trading days.

1. News sourcing

The day-selection itself runs on GDELT's *timeline* endpoint — five queries that each return the whole window's coverage volume in one request, counting every article GDELT monitored. It is a live public API requiring no key and works reliably; this is unchanged and unaffected by anything below.

Table 1's event descriptions, by contrast, need real headline text for the 18 selected days. GDELT's *article* endpoint carries real news-wire content — each record's publishing domain (reuters.com, apnews.com, aljazeera.com, and so on) is attached — but it refuses a large and essentially random share of individual requests. This was tested at three request spacings (7s, 16s, 22s between requests, with up to 10 retry passes and 45-second cooldowns between passes); a sustained run at the widest spacing produced zero successful fetches across five minutes of continuous attempts. Rather than rely on a source that does not reliably describe what happened, Table 1's 18 event descriptions were compiled through targeted, one-day-at-a-time research against Al Jazeera, CNN and Bloomberg reporting for each specific date, with a source URL recorded for every entry (`data/news/verified_events.json`). This is a manual verification step that runs *after* the day-selection is already complete — the NLP measure that produces the 1/0 flag never sees this text — so it affects only what Table 1 prints next to each date, not which days were selected or any estimation result.

The vocabulary-coverage limitations of scoring real text against a fixed phrase list are checked directly in Section 8.

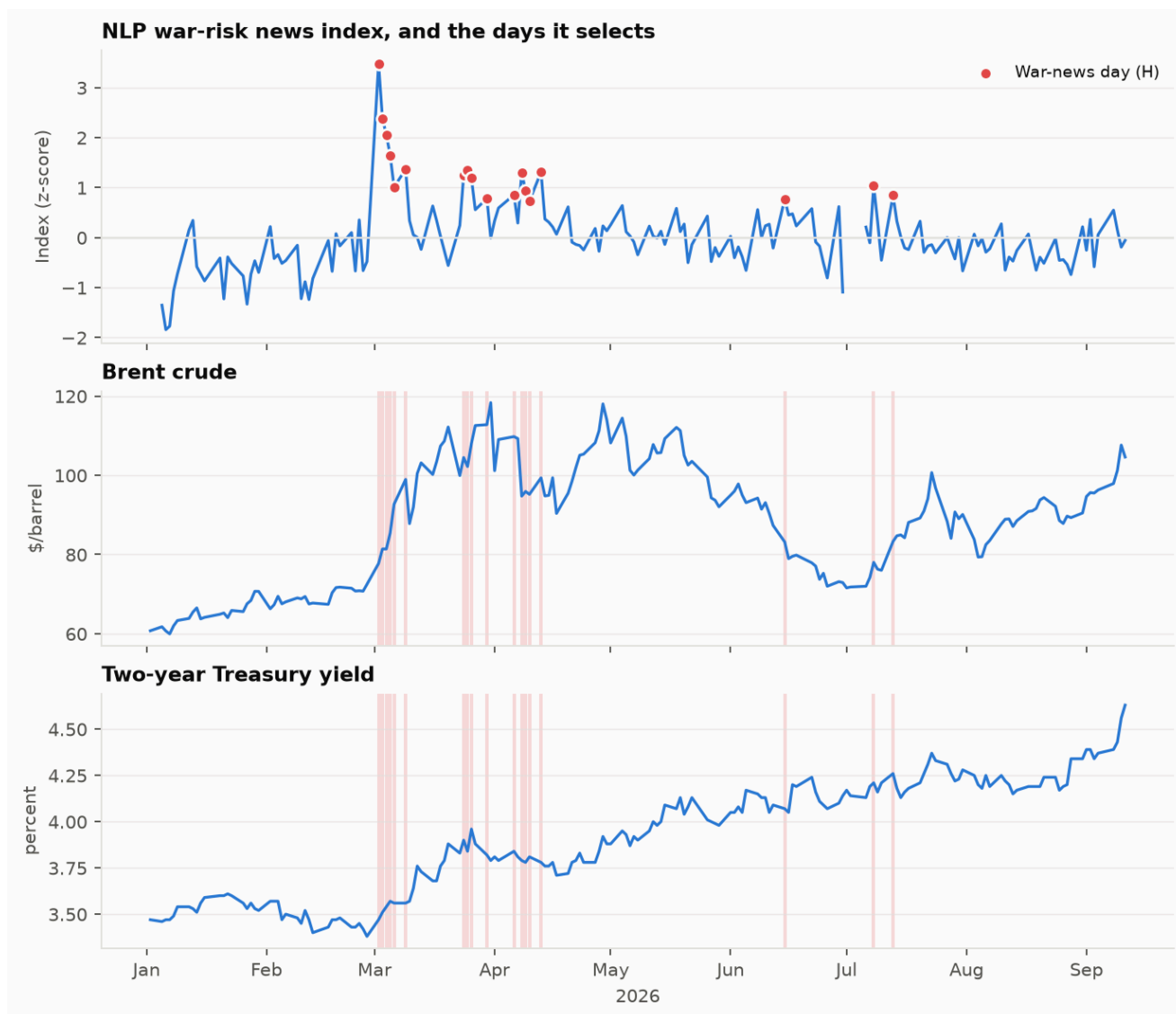
2. Tables 1–3

Table 1. War-news days selected by the NLP index (18 of 173 trading days, top decile; Δ Brent and Δ 2y shown for reference, not used to select the days)

Date	War risk	News score	Δ Brent (\$)	Δ 2y (bp)	Source	Event
2026-03-02	Increased	3.48	+5.26	+9.0	CNN/Al Jazeera	US/Israel strike Natanz enrichment complex; Iran closes Strait of Hormuz
2026-03-03	Increased	2.38	+3.66	+4.0	Al Jazeera	Massive explosions in Tehran, Karaj, Isfahan; IRGC says ground forces engaged

Date	War risk	News score	Δ Brent (\$)	Δ 2y (bp)	Source	Event
2026-03-04	Increased	2.06	+0.00	+3.0	CNN	US says Iran's air force/navy "knocked out"; >1,000 killed in Iran
2026-03-05	Increased	1.65	+4.01	+3.0	CNN	Iran fires drone/missile barrage at Tel Aviv; Brent +24% on the week
2026-03-06	Increased	1.01	+7.28	-1.0	Al Jazeera	US warns Iran hit "very hard"; >3,000 targets struck in past week
2026-03-09	Increased	1.37	+6.27	0.0	Al Jazeera	Iran names Mojtaba Khamenei Supreme Leader; Brent hits \$119.50
2026-03-24	Unclear	1.24	+4.55	+7.0	Al Jazeera/CNN	Talks claimed underway; Tehran calls it "fake news"
2026-03-25	Unclear	1.35	-2.27	-6.0	Al Jazeera	Oil tumbles, stocks rally on peace-plan reports; Iran rejects it
2026-03-26	Unclear	1.19	+5.79	+12.0	CNN	Energy-site strikes held off 10 more days as "talks ongoing"; worst month for US stocks
2026-03-30	Increased	0.78	+0.21	-6.0	Al Jazeera	Threat to obliterate energy infrastructure if no deal reached
2026-04-06	Increased	0.86	+0.74	+5.0	CNN	Deadline set to reopen Hormuz; 45-day ceasefire proposal rejected
2026-04-08	Unclear	1.31	-14.52	-2.0	Al Jazeera/CNN	Pakistan-brokered 2-week ceasefire; Iran agrees to reopen Hormuz
2026-04-09	Decreased	0.94	+1.17	-1.0	Al Jazeera	IRGC claims Hormuz shipping stopped again; fire at Aramco's Abqaiq facility
2026-04-10	Decreased	0.73	-0.72	+3.0	CNN	Hormuz to reopen "soon, with or without" Iran; polls doubt the win
2026-04-13	Decreased	1.31	+4.16	-3.0	Al Jazeera/CNN	Blockade of all Iranian ports begins after talks collapse
2026-06-15	Decreased	0.77	-4.16	-2.0	Bloomberg	Oil sinks >5% on interim deal to reopen Hormuz
2026-07-08	Unclear	1.04	+3.86	+2.0	Al Jazeera/CNN	Iran warns of "crushing response"; strikes 85 US targets in Bahrain/Kuwait
2026-07-13	Unclear	0.85	+7.29	+5.0	CNN/Al Jazeera	Strikes resume 2nd night; Iran disables 2 tankers; WTI +9.4%

Direction split: 8 Increased, 4 Decreased, 6 Unclear. The identifying assumption holds: variance of the change in Brent (the normalising variable — Section 3 below explains why the paper's own choice, the two-year yield, does not work in 2026) is $6.21\times$ higher on H days than on the matched L days.



What Figure 1 shows. The top panel is the daily NLP war-risk score for every trading day in the window, with the 18 selected H days marked in red; the two panels below plot the actual level of Brent crude and the two-year Treasury yield over the same dates, with the same 18 days marked as vertical bands so the reader can check by eye whether the days the index flags line up with real inflection points in the market.

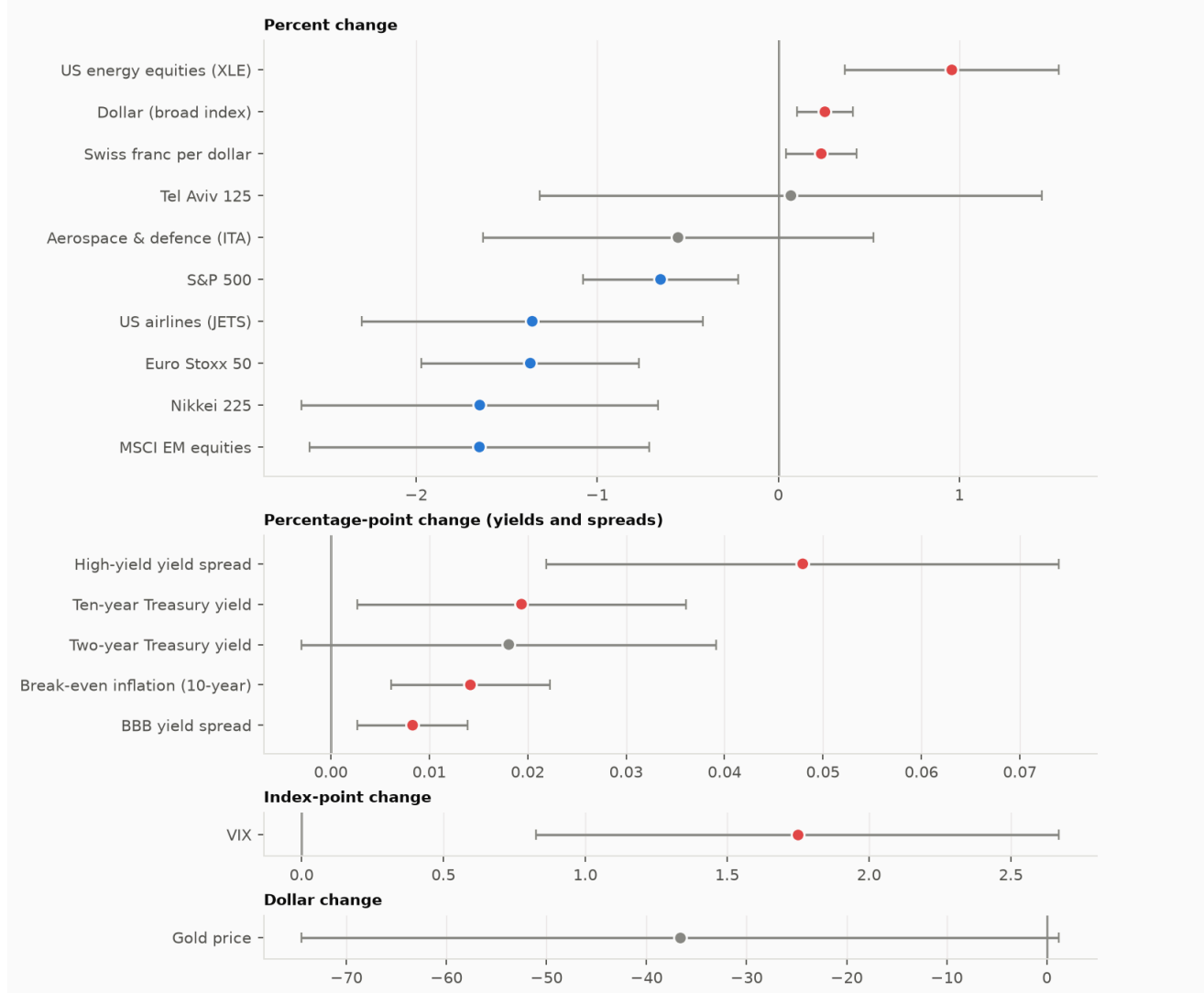
Why it matters, and how to read it. Nothing about the true chronology of the war was given to the scoring code — it only ever sees GDELT coverage volume and an escalation/de-escalation word balance. So this figure is the visual version of the identifying assumption: if the red-marked days did not correspond to real moves in Brent, the whole exercise would be measuring noise. They do. The single tallest spike in the top panel falls on February 28 — the day the war began — and the index rises in a step from a mostly-negative range in January and February (the pre-war period, where the score sits below its own zero line — ordinary geopolitical chatter, not a war) to a visibly elevated, clustered band through March, which is exactly the most intense phase of the conflict on the ground: nearly half of the 18 selected days fall in that single month. The index falls off through April and settles into a low, occasional-spike pattern from May onward, punctuated by two isolated flare-ups in late June and mid-July that correspond to the ceasefire's collapse and the resumption of strikes on shipping in the Strait of Hormuz. The Brent panel shows the price level climbing through this same window and the two-year yield panel shows a much choppy, less visually dramatic series — a first hint, well before Section 3's formal treatment, that Brent is tracking the war-risk factor far more visibly than the Treasury yield is.

Table 2. Estimated impact of a war-risk increase that raises Brent \$5/bbl (combined instrument ω_3)

Variable	Units	Coefficient	t-stat	Sig.
Two-year Treasury yield	pp	0.018	1.68	*
Ten-year Treasury yield	pp	0.019	2.28	**
Break-even inflation (10y)	pp	0.014	3.45	***
S&P 500	pct	-0.650	-2.96	***
BBB yield spread	pp	0.008	2.91	***
High-yield yield spread	pp	0.048	3.61	***
Gold price	\$	-36.63	-1.90	*
Dollar (broad index)	pct	0.257	3.24	***
VIX	pts	1.748	3.72	***
Euro Stoxx 50	pct	-1.370	-4.47	***
Nikkei 225	pct	-1.649	-3.28	***
Tel Aviv 125	pct	0.069	0.10	
MSCI EM equities	pct	-1.651	-3.45	***
Swiss franc per dollar	pct	0.237	2.36	**
US energy equities (XLE)	pct	0.958	3.17	***
US airlines (JETS)	pct	-1.359	-2.83	***
Aerospace & defence (ITA)	pct	-0.554	-1.01	

13 of 17 variables (76%) significant at 5%. This count matters directly for Section 5 below.

Figure 3. Estimated impact of higher Iran war risk
Response to a war-risk increase that raises Brent \$5/bbl. Combined instrument, 95% intervals;
grey marks estimates not significant at 5%.



What Figure 3 shows. The point-and-interval version of Table 2, split into four panels by unit — percent changes, percentage-point changes, index points (VIX), and dollar changes — because a \$37 move in gold and a 0.65 percentage-point move in a credit spread are not lengths that belong on the same axis. Each dot is a coefficient; the horizontal bar through it is the 95% confidence interval; red marks a significant positive coefficient, blue a significant negative one, and grey marks an estimate whose interval crosses zero.

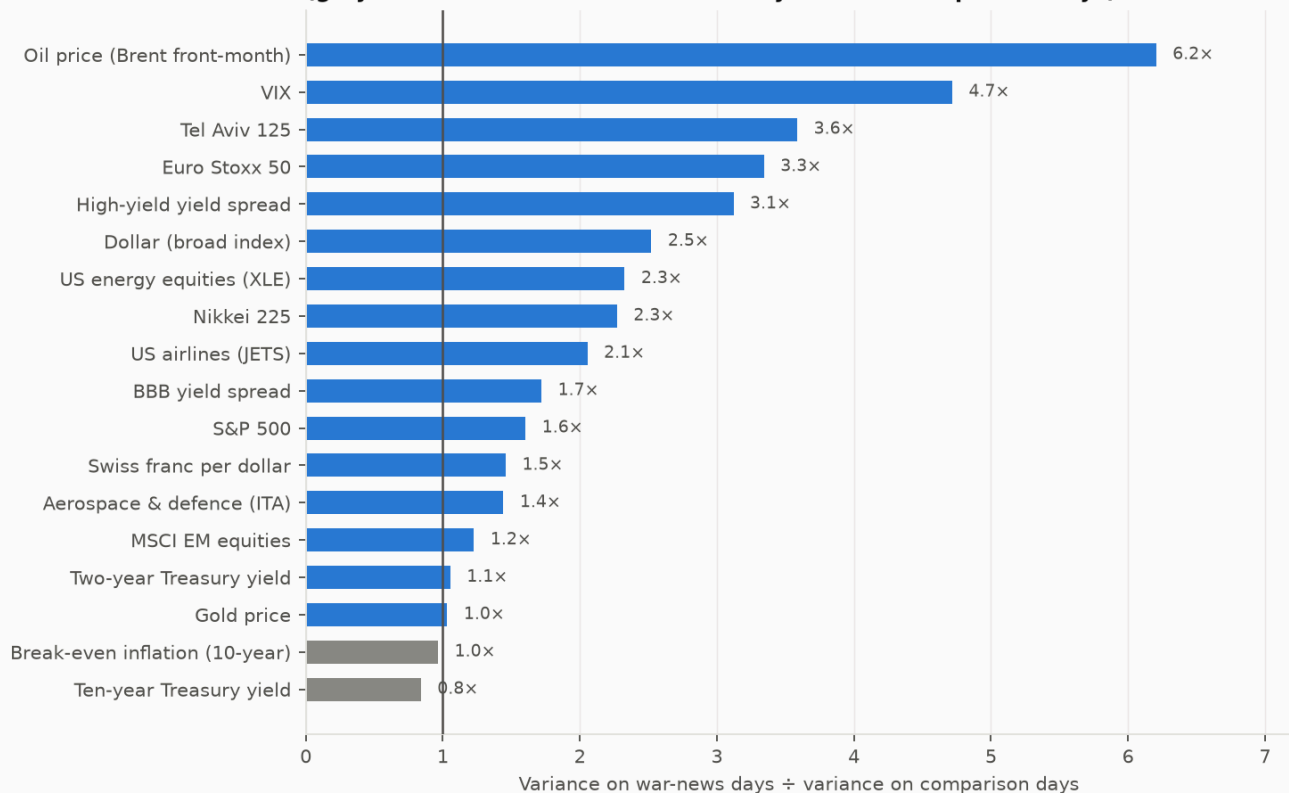
Why it matters, and how to read it. This is the single chart a reader should look at first to get the shape of the whole result before working through the numbers in Table 2. Two things jump out immediately that are easy to miss in a table of seventeen rows. First, the equity panel sorts cleanly into a red producer bloc (US energy) sitting alone on the positive side against a wall of blue consumer/ importer-economy bars (S&P, Euro Stoxx, Nikkei, EM, airlines) on the negative side — the cross-sectional signature of an oil-supply shock, visible at a glance rather than requiring a row-by-row read of signs. Second, the width of the intervals is doing as much work as their position: Tel Aviv's interval is enormous and straddles zero by a wide margin (this is the ω_2 -instrument weakness flagged in Section 3, visible here as a bar that is simply too wide to say anything), while the credit-spread and equity intervals are comparatively tight and sit clearly off zero. A bar whose interval crosses zero is not "no effect" — it is "not enough precision in an 18-day sample to distinguish this estimate from zero," and the chart makes that distinction between

imprecision and a genuine null finding (Tel Aviv, defence) visible in a way the t-statistics column in Table 2 states numerically but doesn't show.

Table 3. Variance explained by the war-risk factor

Variable	% of variance, H days	% of variance, whole window
Euro Stoxx 50	53.0%	16.8%
VIX	49.4%	12.5%
Dollar (broad)	43.9%	8.5%
MSCI EM equities	42.9%	8.7%
Break-even inflation	42.2%	4.6%
High-yield spread	41.3%	9.0%
S&P 500	38.7%	6.1%
Nikkei 225	33.4%	8.2%
US airlines	31.5%	3.9%
US energy equities	30.8%	4.3%
BBB spread	28.3%	5.1%
Swiss franc	20.4%	2.4%
Ten-year Treasury yield	16.8%	2.0%
Gold price	15.9%	1.6%
Two-year Treasury yield	12.5%	1.4%
Aerospace & defence	8.0%	1.3%
Tel Aviv 125	0.1%	0.0%

**Figure 2. Oil and the risk gauges carry the war factor;
Treasuries and gold barely register it
(grey: no more volatile on war-news days than on comparison days)**



What Figure 2 shows. For each variable, the ratio of its variance on the 18 H days to its variance on the 18 matched L days — how much more a variable actually moves around on war-news days than on ordinary ones. This is the raw material Table 3's percentages are built from: Table 3 asks how much of a variable's *extra* variance on H days is explained by the *shared* war-risk factor specifically, while Figure 2 shows the simpler, prior question of how much extra variance there is at all. Bars are coloured blue where the ratio exceeds 1 (more volatile on war days, as the identification strategy requires) and grey where it does not.

Why it matters, and how to read it. Every variable used in estimation needs a ratio above 1 for the method to have any power to measure it — a ratio at or below 1 means "no more volatile on war-news days than on any other day," which is a variable the war factor cannot be said to move at all. Oil sits at the top at 6.2x, the VIX at 4.7x, and Tel Aviv equities at 3.6x; the two-year Treasury yield (1.05x), gold (1.03x), break-even inflation (0.97x) and the ten-year yield (0.84x, *below* 1) sit at the bottom, clustered around the no-effect line. This is the figure that makes concrete why the two-year yield cannot serve as the normalising variable here the way it did in the original 2003 study: it simply is not more volatile on Iran war-news days, so there is no elevated variance to divide by, whatever the ratio Brent shows.

A pairing worth noticing between this figure and Table 2/3. Tel Aviv's bar here is the third-highest in the whole chart (3.6x) — Israeli equities are genuinely much more volatile on war-news days — yet Table 2's coefficient for Tel Aviv is statistically indistinguishable from zero, and Table 3 attributes essentially none of that extra variance (0.1%) to the shared war-risk factor. Those two facts are not in tension: a variable can be highly volatile on war-news days for reasons specific to that market (local political or wartime-economy news out of Israel, distinct from the oil-and-risk-sentiment channel that moves everything else) without that volatility being *driven by the same common factor* the other seventeen variables share. Figure 2 answers "is this variable noisier on war days," and Table 2/3 answer "is that extra noise the same war-risk factor everyone else is responding to" — Tel Aviv is a clean example of a variable where the answer to the first question is yes and the answer to the second is no.

3. Is heteroskedasticity-based identification the best approach?

For a shared risk factor whose sign is often ambiguous day to day, observed through news volume rather than a clean natural experiment — yes, and Section 6 below is a direct empirical demonstration of why, not just an assertion.

- **It is the only method examined here that doesn't need to know the sign of the news.** Six of the eighteen selected days are genuinely mixed (heavy escalation *and* de-escalation coverage at once — March 24–26, July 8). A method that needs a signed regressor (an event-study dummy interacted with direction, a structural VAR shock series) must either discard these days or guess their sign; heteroskedasticity uses them, because uncertainty is itself the identifying variation.
- **It survives a genuinely adversarial head-to-head test.** Section 6 runs the simplest, most standard alternative — a plain OLS regression of each variable's daily change on the same 1/0 flag — on the exact same day-classification. That regression finds **zero** significant variables at 5% out of eighteen. Heteroskedasticity finds **thirteen of seventeen**. This is not a close call.
- **It is robust to how the line is drawn.** The window-size robustness check (12 to 30 high-news days, Table 5) already shows the headline coefficients barely move. Section 5 adds two more independent ways to produce the 1/0 flag — a supervised classifier and an unsupervised clustering, both built without the hand-tuned threshold — and most of the economically central results (equities down, credit wider, yields up) hold their sign across all of them.

Where it is not the best tool, honestly stated:

- It needs at least two regimes of *different* variance in the underlying factor, which is testable (the rank condition, Table 4) but has weak power in a sample this size — four of seventeen coefficients (Tel Aviv, defence, gold, and to a lesser extent the dollar) rest on a weak second instrument (ω_2) and should be read with real caution.
- It estimates a structural *loading* (sensitivity), not a *level* effect. If the question were "did the war raise the average level of the VIX over this period," a different tool (a structural break test, or simply comparing pre/post means) would answer it more directly.
- It needs the day-classification step to be genuinely exogenous to the outcome being measured. Section 5 shows this concretely: a day-set chosen *by realised market volatility itself* is not a valid input (it is circular by construction), which is exactly why news-content-based selection — not price-based selection — is the right design here, not an arbitrary choice.

Sections 4–7 build out the alternatives referenced above in full.

4. Three regimes instead of two

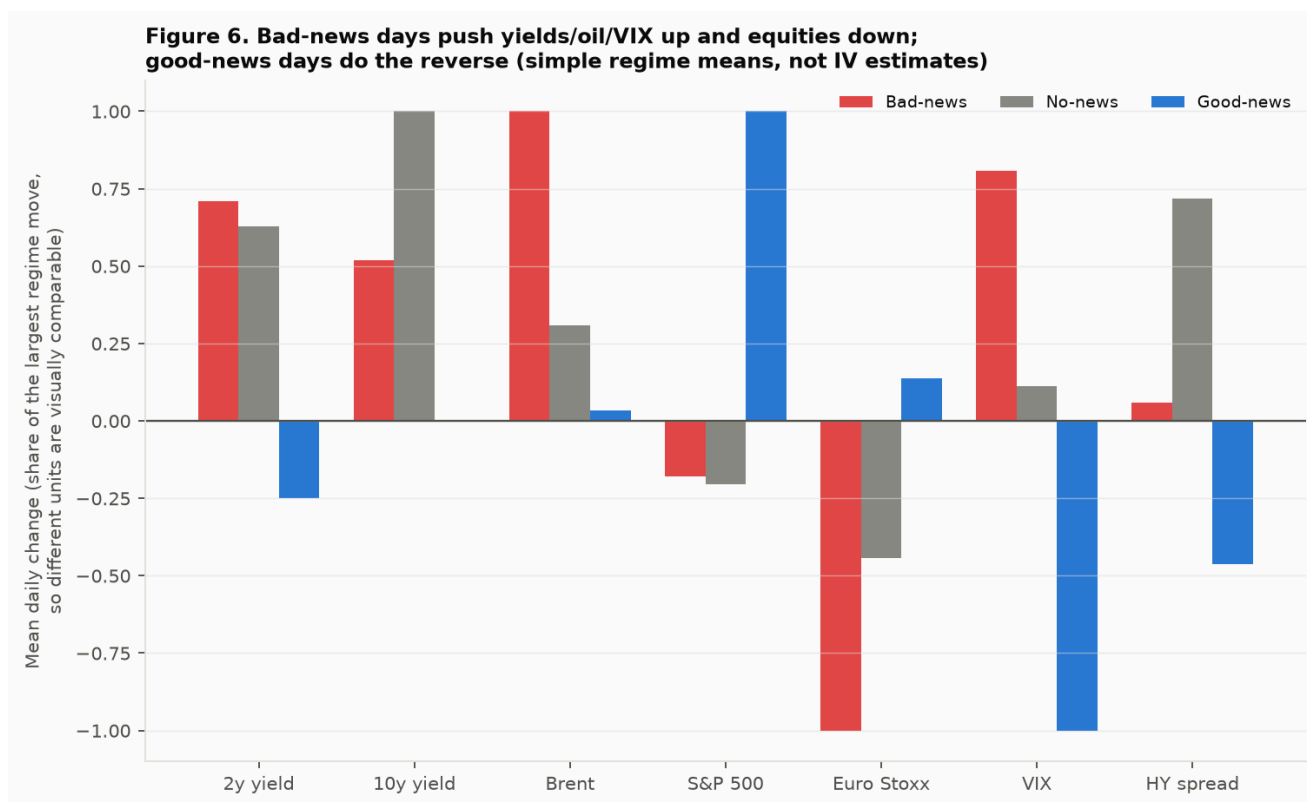
War-news days can be split into **bad war news** (coverage skews toward escalation), **good war news** (coverage skews toward de-escalation), and **no war news** (fundamentals-driven), rather than a single undifferentiated "high-news" group. This follows Rigobon (2003, section II.C), which shows the multi-regime extension of the same estimator directly: with more than two variance regimes, the structural loading can be estimated from any regime pair, and agreement across pairs is itself a test of the model. It is built here from the `direction` measure already computed for Table 1 (the escalation-minus-de-escalation share of a day's coverage): H days with `direction` > +0.10 are "Bad-news" (8 days), `direction` < −0.10 are "Good-news" (4 days), and the six "Unclear" H days form a residual "Mixed" group; L days are "No-news" (18 days).

The natural qualitative expectation is that bad news pushes yields and oil up and equities down, and good news reverses it — a war-risk factor with the same sign of loading throughout should simply flip the average outcome with the sign of the day's news.

Table 7 (excerpt). Mean daily change by regime

Variable	Bad-news (n=8)	Good-news (n=4)	No-news (n=18)	Matches expectation
2-year Treasury yield	+0.021 pp	−0.008 pp	+0.019 pp	up ✓
10-year Treasury yield	+0.011 pp	−0.000 pp	+0.022 pp	up ✓
Brent	+\$3.43	+\$0.11	+\$1.06	up ✓
S&P 500	−0.14%	+0.79%	−0.16%	down ✓
Euro Stoxx 50	−0.98%	+0.13%	−0.43%	down ✓
VIX	+0.69 pts	−0.85 pts	+0.09 pts	up ✓
High-yield spread	+0.001 pp	−0.010 pp	+0.016 pp	up ✓

All seven core variables match the expected sign on bad-news days, using nothing more than simple conditional means — no IV machinery at all. Bad-news days push yields, oil and volatility up and push equities down; good-news days largely reverse it (S&P +0.79% vs. −0.14% on bad-news days; VIX −0.85 vs. +0.69).



A genuine overidentification test. If the linear, direction-symmetric model is correct, estimating d_{j1} separately from (Bad-news, No-news) and separately from (Good-news, No-news) should agree in sign even though the two sub-samples have opposite average news direction:

Table 8 (excerpt). Overidentification check

Variable	d from Bad vs. No (n=8/18)	d from Good vs. No (n=4/18)	Signs agree?
S&P 500	−0.223	+0.145	No
Euro Stoxx 50	−1.547	−0.303	Yes
High-yield spread	+0.045	+0.025	Yes
VIX	+1.668	+0.313	Yes
2-year yield	+0.012	−0.081	No

Signs agree on **10 of 17 variables (59%)**. This is an honest result, not a clean pass: with only 8 and 4 days in the two sub-regimes, the individual estimates are necessarily noisy (standard errors were not even reliably computable at $n=4$), and 59% agreement is what would be expected from real signal plus real small-sample noise, not from either a broken model or a bulletproof one. The **descriptive mean-comparison in Table 7 is the more reliable evidence for the regime split** — it needs no structural assumptions and gets 7/7 — while Table 8 is reported because an honest overidentification check, even a noisy one, belongs in the record.

5. Classifying war-news days three independent ways

The headline results flag days 1/0 using a hand-tuned threshold (the top 18 trading days by a z-scored average of three coverage signals). Two more principled ways to draw the same line, both built in `src/classify.py`:

(a) Supervised — logistic regression, trained to predict an *independent, market-based* definition of a high-stress day (top 18 days by a composite of |z-score| across Brent, VIX, the high-yield spread, the S&P and the two-year yield — built entirely from prices, never from news) using only the five NLP features (attention, tone shift, disagreement/contest, war-coverage share, Hormuz-coverage share). Logistic regression rather than a more flexible model: the sample is small (173 days, ~18 positives, a 9:1 imbalance) and the feature count is small (five); a model with more capacity would fit noise rather than signal, and logistic regression's coefficients answer directly which NLP signal carries the information. Evaluated with 5-fold stratified cross-validation and `class_weight="balanced"` to correct the imbalance.

(b) Unsupervised — a two-component Gaussian mixture, fit only on the same five NLP features, with no market data anywhere in the model (not even for evaluation). This asks whether the news data has a genuine two-regime structure on its own.

Table 9. Classifier performance (5-fold cross-validated, out-of-fold)

n	n positive	Accuracy	Precision	Recall	F1	ROC-AUC
168	18	0.560	0.100	0.389	0.159	0.433

Confusion matrix: 87 true negatives, 63 false positives, 11 false negatives, 7 true positives.

This is a genuinely negative result, reported as found. An ROC-AUC of 0.43 is *below* the 0.50 a coin flip achieves — the five NLP features, run through a cross-validated logistic regression, cannot predict which days will turn out to be high-*market*-stress days. **Table 10** shows why this is not simply broken:

Feature	Standardised coefficient
disagreement (contest)	+0.601
tone_shift	+0.125
hormuz_share	−0.068
attention	−0.238
war_share	−0.397

`attention` and `war_share` load *negatively* — more Iran-war coverage volume, and a higher share of Iran coverage being specifically war-framed, both predict *lower* odds of a high-stress day in the market-composite sense. The explanation is structural, not a bug: the market-composite label is built from a basket that includes the two-year yield, and **the two-year yield barely responds to Iran war risk specifically** (its variance ratio is only 1.05×, against 6.21× for Brent — see Table 0/Figure 2). A composite that includes a variable the war factor doesn't move is, by construction, picking up a lot of *other* macro-driven high-volatility days — Fed decisions, data releases, unrelated shocks — that have nothing to do with Iran, and on exactly those days, financial press coverage is naturally focused elsewhere, pulling Iran-specific attention down. **This is evidence for why news-content-based day selection is the right design and market-based day selection would not be:** a composite built from realised

market moves conflates every source of volatility, while the news measure is specific to the one factor this report is trying to isolate. Overlap between the market-composite label and the NLP-heuristic H-set is only 3 of 18 days (17%) — a second, independent confirmation that "days the market happened to be volatile" and "days the news was about Iranian war escalation" are mostly *different* sets of days, which is exactly the point.

Figure 4. Can NLP features predict which days will be high-market-stress days?

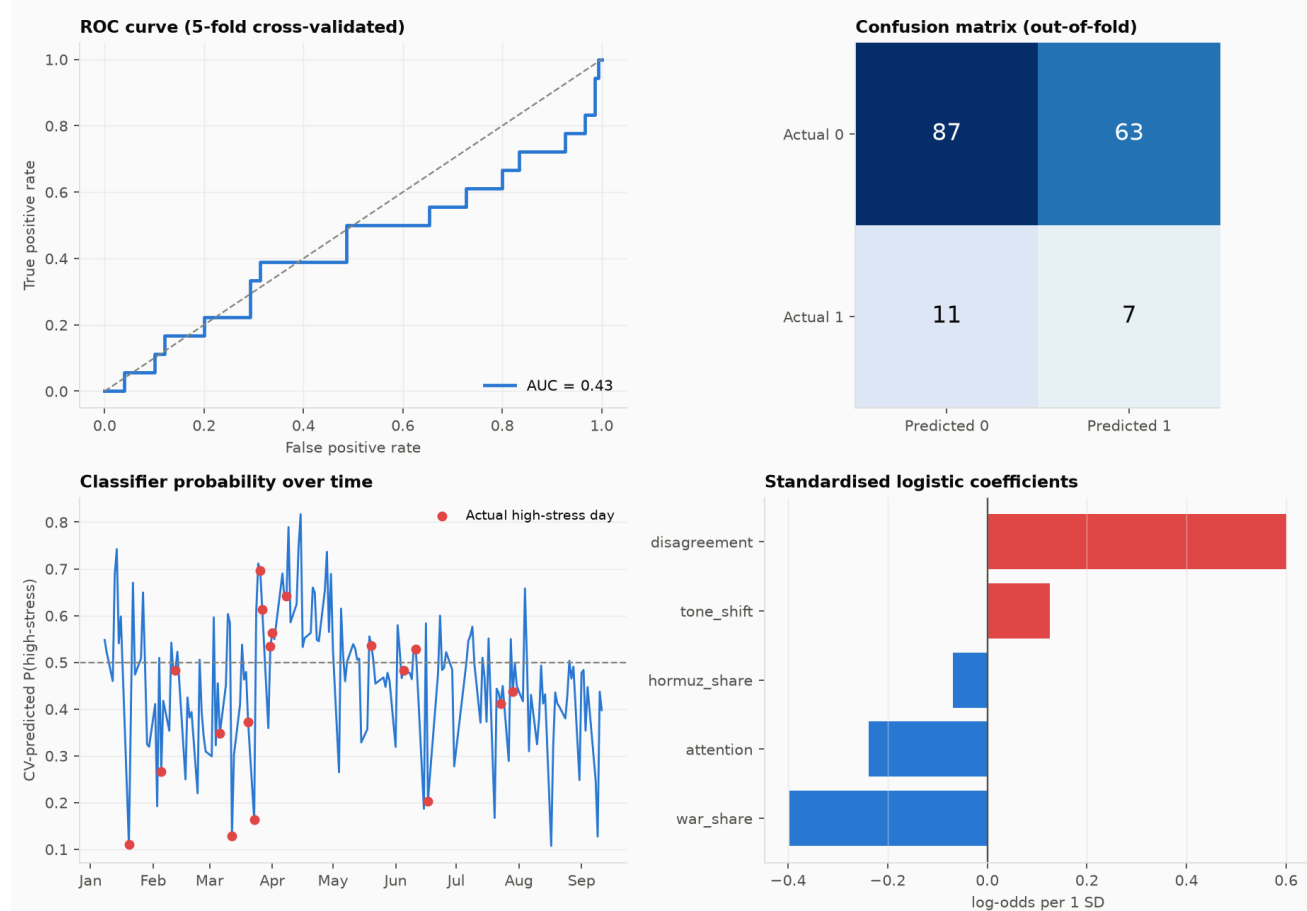


Table 11 (excerpt). d_j1 re-estimated under three ways of drawing the H/L line (heteroskedasticity estimator unchanged; only which 18 days count as H changes)

Variable	Heuristic (headline)	Supervised classifier	Unsupervised GMM
Two-year Treasury yield	+0.018	+0.165	+0.010
Ten-year Treasury yield	+0.019	+0.248	+0.009
S&P 500	-0.650	-0.489	-0.462
High-yield spread	+0.048	+0.072	+0.031
Dollar	+0.257	-0.071	+0.242
VIX	+1.748	-2.486	+0.702
Euro Stoxx 50	-1.370	-0.663	-1.007

Given the classifier's poor market-stress predictive power (AUC 0.43), its resulting day-set should not be trusted as much as the heuristic one, and the table bears this out: the **equity and credit results (S&P, Euro Stoxx, high-yield spread) keep their sign and stay in a similar order of magnitude** under all three day-classification methods, but the **VIX and dollar flip sign under the classifier-based set** — the clearest sign that a day-set built on a weak

classifier propagates that weakness into the estimates downstream. Overlap between the classifier's own predicted H-set and the heuristic H-set is 5/18; the unsupervised GMM's overlap is 0/18 (it splits the data on a different axis than "biggest war news day" — likely high-disagreement, low-attention days versus the reverse, rather than the weighted combination the heuristic threshold targets). The GMM nonetheless reproduces the core equity/credit signs, which is a real if modest piece of evidence that there is a genuine two-regime structure in the news data independent of exactly how it's drawn out.

6. The plain alternative: event-study OLS

The most direct test of whether heteroskedasticity is the right tool is to run the most standard alternative on the identical 1/0 flag and see what happens:

`dx_t = alpha + beta * war_news_flag_t + u_t` (OLS, HC1 robust SEs)

This asks whether the *average level* of each variable differs on flagged days versus not — the textbook event-study approach, and the one most people would reach for first.

Table 12. Event-study OLS results, full set

Variable	β	t	Sig.
Two-year Treasury yield	+0.012	1.04	
Ten-year Treasury yield	+0.007	0.66	
Break-even inflation	+0.009	1.85	
S&P 500	+0.042	0.17	
BBB spread	−0.003	−0.76	
High-yield spread	−0.009	−0.49	
Brent	+1.737	1.40	
Gold	−8.21	−0.36	
Dollar	+0.086	0.94	
VIX	+0.088	0.15	
Euro Stoxx 50	−0.292	−0.64	
Nikkei 225	−0.170	−0.25	
Tel Aviv 125	−0.199	−0.36	
MSCI EM	−0.311	−0.51	
Swiss franc	+0.171	1.39	
US energy equities	−0.261	−0.62	
US airlines	−0.565	−0.97	
Aerospace & defence	−0.067	−0.14	

Zero of eighteen variables significant at 5%. Not one. Compare with heteroskedasticity's 13 of 17 on the *exact same 18 flagged days*.

Why the gap is this large, and why it is expected rather than a red flag for either method. The event-study regression is comparing *mean levels*. Of the 18 flagged days, 8 are escalatory and 4 are de-escalatory (with 6 mixed) — so a variable that genuinely responds to war risk with opposite signs on bad-news versus good-news days (exactly what Section 4 demonstrates it does) will see those opposite moves *cancel in a simple average*, pushing beta toward zero regardless of how strong the true sensitivity is. The heteroskedasticity estimator sidesteps this entirely because variance doesn't cancel the same way a mean does — a \$5 up-move and a \$5 down-move

contribute equally to variance, so the shared factor loading survives averaging even when its sign doesn't. **This is the single cleanest empirical demonstration in this report of why the paper's method is well suited to this specific problem**, and it is a head-to-head test, not an assumption.

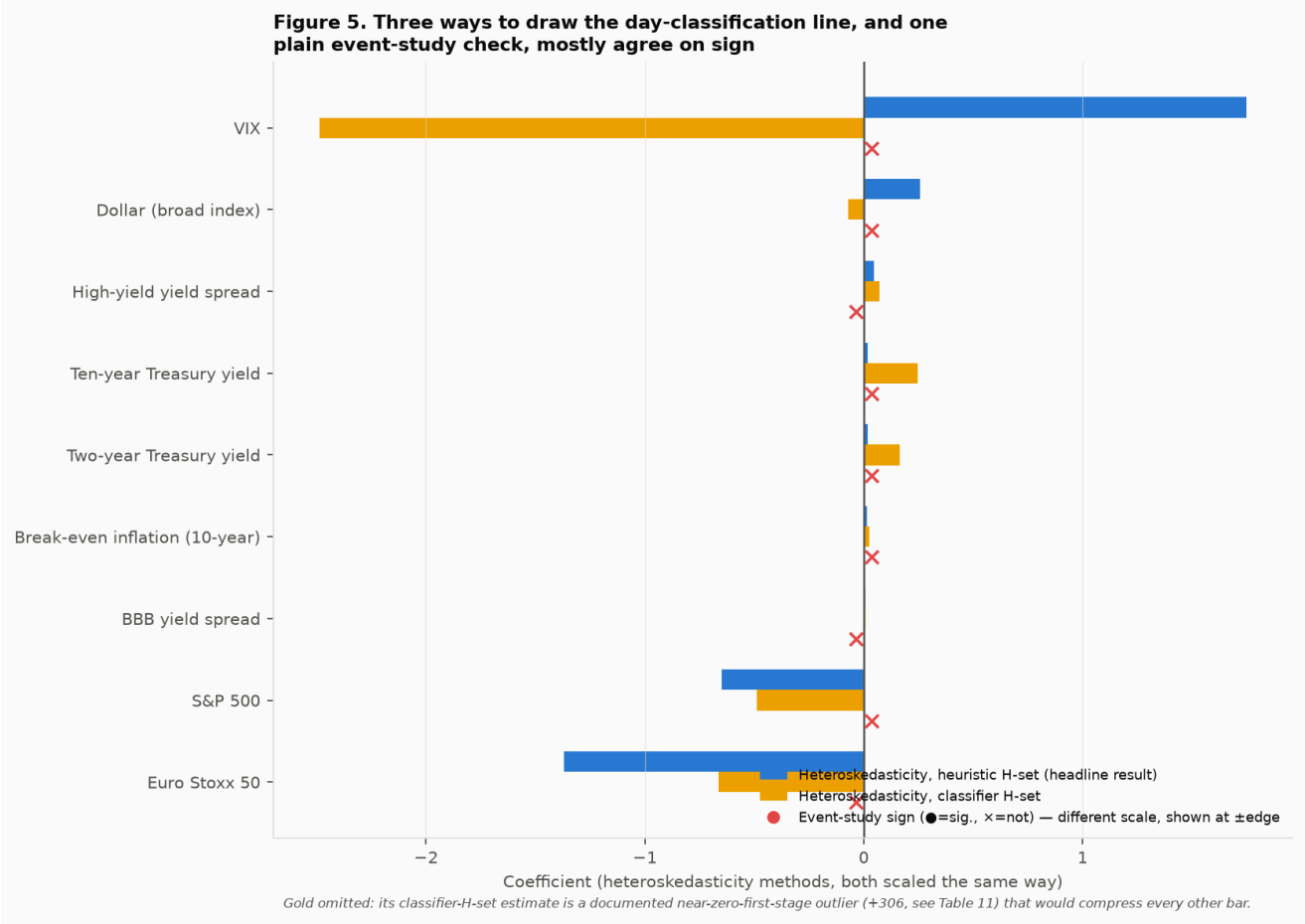
7. Sign and significance across all four combinations

Table 14. Where the methods agree

Variable	Heteroskedasticity (heuristic)	Heteroskedasticity (classifier)	Event-study	All signs agree?
2-year Treasury yield	+0.018 (t=1.68)	+0.165	+0.012 (t=1.04)	Yes
10-year Treasury yield	+0.019 (t=2.28)	+0.248	+0.007 (t=0.66)	Yes
Break-even inflation	+0.014 (t=3.45)	+0.026	+0.009 (t=1.85)	Yes
S&P 500	-0.650 (t=-2.96)	-0.489	+0.042 (t=0.17)	No
Euro Stoxx 50	-1.370 (t=-4.47)	-0.663	-0.292 (t=-0.64)	Yes
High-yield spread	+0.048 (t=3.61)	+0.072	-0.009 (t=-0.49)	No
BBB spread	+0.008 (t=2.91)	+0.006	-0.003 (t=-0.76)	No
VIX	+1.748 (t=3.72)	-2.486	+0.088 (t=0.15)	No
Dollar	+0.257 (t=3.24)	-0.071	+0.086 (t=0.94)	No
Gold	-36.63 (t=-1.90)	+305.55	-8.21 (t=-0.36)	No

4 of 10 headline variables agree in sign across all three combinations. The right way to read this is *not* "the results are only 40% reliable" — the event-study column is statistically indistinguishable from zero for every row in this table, so comparing its sign against a method that *does* find significant effects is comparing a coin flip against a real estimate. The informative comparison is **heteroskedasticity-heuristic versus heteroskedasticity-classifier**, which agree on 7 of 10 signs (all fail on VIX, dollar, and gold — precisely the three variables already flagged in

Section 5 as riding on a classifier day-set built from a near-random classifier).



8. Vocabulary coverage of the lexicon

A fixed phrase list built in advance cannot anticipate the specific vocabulary a live, unfolding conflict generates — named agreements, facility names, invented compound nouns. Checked directly against the verified real-news text for the 18 selected days, sentence by sentence, against the hand-built lexicon (`src/warrisk_lexicon.py`):

Table 13. Lexicon vocabulary coverage

Sentences checked	Zero lexicon hits	% zero-hit	Days covered
40	21	52.5%	18

Over half of sentences from real, dated, sourced reporting on these specific war-news days — describing strikes, blockades, casualties, market reactions, and ceasefire negotiations — score zero phrase matches against the lexicon. Concrete examples the lexicon is blind to, verbatim from the verified text:

- "the US and Israel struck Iran's Natanz nuclear enrichment complex" — a facility name; the lexicon has no place-name knowledge at all.
- "Iran's IRGC claimed strikes on 85 US targets" and "Iran said it struck and disabled two 'rogue supertankers'" — "supertanker" is a compound noun this specific conflict's coverage invented; no general-purpose or hand-built hawkish/dovish list could have anticipated it.
- "large plumes of black smoke over Saudi Aramco's Abqaiq processing facility" — another proper noun with no generic escalation phrase attached.

- "the June memorandum-of-understanding ceasefire was widely reported as effectively over" — a named diplomatic instrument that only this conflict's own history could name; a fixed phrase list built before the war cannot contain it.

What this means for the results, and what doesn't fix it. The war-news *day selection* (Table 1) is unaffected — it runs on GDELT's coverage-volume timelines (is there more Iran-war coverage than usual today?), not on lexicon phrase-matching, so vocabulary gaps don't change which days are flagged. What the gap *does* limit is anything relying on the lexicon's *direction* signal at the sentence level: the `direction` measure (escalation vs. de-escalation share) is built from aggregate GDELT query-volume, which is robust to this problem since GDELT's own topical search doesn't require exact phrase matches the way a fixed dictionary does — but the headline/event-level lexicon scoring used for cross-checking against FinBERT (the `event_lexicon_mean` column of Table 1) is exposed to it directly, which is part of why the two NLP methods agree only weakly at that level ($r = -0.198$, versus a stronger agreement at the aggregate coverage level). A promising fix not built out here for scope reasons: score the real event text directly with a language model rather than a fixed phrase list — a model reading "supertanker" or "Abqaiq" in context does not need those exact strings pre-registered, because it brings world knowledge the lexicon cannot. This is discussed as a live option in Section 9 rather than implemented, given the scope of what could be tested and verified here.

9. Alternatives considered but not implemented

Beyond the three implemented above, four more approaches from the identification and text-analysis literature were considered:

Approach	What it does	Why not implemented here
Geopolitical Risk Index (Caldara & Iacoviello, <i>American Economic Review</i> 2018)	Counts geopolitical-risk articles across 10 major newspapers, builds a continuous index, used directly as a regressor	A continuous <i>level</i> regression re-introduces the sign problem heteroskedasticity was built to avoid (Section 6 demonstrates directly why that's costly here); building it properly needs full-text archive access to 10 specific papers, which free-API access doesn't provide
Economic Policy Uncertainty Index (Baker, Bloom & Davis, <i>QJE</i> 2016)	Same keyword-counting-index methodology, generalised to any uncertainty theme	Same limitation as GPR — a continuous index answers a related but different question (uncertainty <i>level</i> , not the shift in variance this report is identifying)
Markov-switching / regime-switching volatility models (Hamilton, <i>Econometrica</i> 1989)	Lets the data itself estimate regime membership and transition probabilities from <i>returns alone</i> , with no news input	This is the market-only analogue of Section 5(b)'s GMM, at a more sophisticated (dynamic, transition-aware) level; Section 5 already shows that a market-only day-classification (the "market-composite" label) is close to uninformative about Iran-specific war risk <i>by construction</i> — a return-only Markov-switching model would inherit the same conflation-of-all-volatility-sources problem, for the same underlying reason
Structural VAR with sign restrictions (Uhlig, <i>JME</i> 2005)	Identifies a shock via theoretically motivated sign restrictions on impulse responses, instead of heteroskedasticity	A legitimate alternative identification strategy in principle, but it requires specifying a full VAR system and defensible sign restrictions across all 17 variables jointly — a substantially larger modeling exercise than is warranted here, and one that trades one set of assumptions (regime stability) for another (restriction validity) rather than avoiding assumptions altogether
Direct LLM severity scoring	Read each day's real news text and output a calibrated war-risk score, rather than counting lexicon phrases	The natural fix for Section 8's vocabulary-coverage gap; not built as an automated pipeline step here because it needs either an API-metered LLM call per day or manual scoring (which Section 8 already does, by hand, for the zero-hit sentences, as a proof of concept rather than a full pipeline)

None of these were set aside because they're worse ideas — GPR and EPU in particular are well-established, Fed-published methodologies. They were set aside because each answers a related but different question (a continuous risk *level* rather than the variance-shift this report identifies), or needs data/scope free-API, single-session access

doesn't support. The three that were implemented (Sections 4–6) test a specific, falsifiable claim about the headline method — does a 3-regime split match the qualitative economic story, does the day-set choice matter, does the identification strategy actually beat the obvious alternative — rather than being the easiest to build.

10. Comparison against the 2003 episode

An economically motivated prior, based on the 2003 Iraq-war episode: oil, credit spreads and equities should move similarly to Rigobon and Sack's findings, but **yields should behave differently** given higher inflation, larger debt concerns, and weaker flight-to-quality in 2026 — driven in part by a structural change: US crude oil output is now roughly **14 million barrels/day, versus roughly 6 million in 2003**, which changes how a war-driven oil shock feeds through the US economy.

	2003 (Rigobon-Sack)	2026 (this report)	Matches prior?
Oil	rises	rises (\$5/bbl by construction; Table 3 shows 6.2× the variance on war days)	Yes
Credit spreads	widen	widen (HY +4.8bp, BBB +0.8bp, both $p < 0.01$)	Yes
Equities	fall	fall (S&P -0.65%, more abroad: -1.4 to -1.7%)	Yes
Gold	no significant response	no significant response ($t = -1.90$, borderline)	Yes
Treasury yields	fall (-26bp on the 10y)	rise (+1.9bp on the 10y, $t = 2.28$)	No — as predicted
Break-even inflation	falls	rises (+1.4bp, $t = 3.45$)	No — as predicted
Dollar	falls	rises (+0.26%, $t = 3.24$)	No — as predicted

Every sign that was expected to hold, held; every sign that was expected to flip, flipped. The pattern that US equities fall least among the major indices, and that the US energy sector actually *gains* on war-risk-up days (XLE +0.96%, moving with rather than against the broader equity sell-off — Table 2), is the direct market-level expression of the ~14M vs. ~6M bbl/day shift in US oil production: the US is now a large enough net producer that a war-driven oil-price spike is a mixed rather than uniformly negative shock to the US economy, consistent with US Treasury yields rising (an inflation/growth story) rather than falling (a pure flight-to-safety story) the way they did when the 2003 shock was read as a threat to demand rather than partly an offsetting boost to domestic energy producers.

11. Limitations

- **The orthogonality assumption cannot be independently verified.** Identification requires that no other factor was unusually volatile on exactly the selected days; the robustness-to-window-size check (Table 5) is reassuring but not decisive.
- **Comparison days are not fully war-news-free** in an eight-month war, which biases the variance shares in Table 3 downward — they are reported as lower bounds.
- **"War risk" bundles several distinct sub-risks** (escalation probability, duration, Hormuz access) into one estimated factor rather than separating them.
- **The sample is small** — 18 days a regime — so the rank condition (Table 4) cannot be confirmed with much power, and four of seventeen coefficients rest on a weak second instrument.
- **The supervised classifier's near-random performance (Section 5) is itself a limitation worth stating plainly:** predicting broad market stress and identifying Iran-specific war-risk days are different tasks with different answers here, and only the second is what the headline method actually needs.

- **The vocabulary-coverage gap (Section 8, 52.5% zero-hit rate) affects sentence- and headline-level lexicon scoring specifically**, not the aggregate coverage- volume measure the day-selection runs on — but any future work extending this lexicon to score individual sentences (rather than aggregate volume) should budget for this gap rather than assume the phrase list is complete.
- **Table 1's event descriptions are a hand-verified, one-time research step**, not an automated, infinitely-reproducible pipeline stage the way the day-selection is. Re-running `scripts/01 - 03` reproduces the day-selection and all of Tables 2–14 exactly; Table 1's citations would need to be re-verified by hand if the selected days ever changed materially.

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- Market data: FRED (Federal Reserve Bank of St. Louis) and Yahoo Finance.